Chapter 7, Problem 36P

Problem

Solve for v in the following differential equations, subject to the stated initial condition.

(a) $dv/dt + v = u(t), \qquad v(0) = 0$

(b) $2\,dv/dt - v = 3u(t), \qquad v(0) = -6$

Step-by-step solution

Step 1 of 4

(a)

Consider the differential equation.

$$\frac{dv}{dt} + v = u(t)$$

Simplify further.

$$\frac{dv}{dt} = u - v$$

$$\frac{dv}{u-v} = dt$$

Step 2 of 4

Integrate on both sides with respect to t.

$$\int_{v(0)}^{v(t)} \frac{dv}{u-v} = \int_0^t dt$$

$$-\ln(u-v)\Big|_{v(0)}^{v(t)} = t\Big|_0^t$$

$$\ln(u - v(0)) - \ln(u - v(t)) = t - 0$$

$$\ln\left(\frac{u - v(0)}{u - v(t)}\right) = t$$

Substitute 0 for $v(0)$.

$$\ln\left(\frac{u(t) - 0}{u(t) - v(t)}\right) = t$$

$$\frac{u(t) - 0}{u(t) - v(t)} = e^t$$

$$u(t) - v(t) = e^{-t}u(t)$$

$$v(t) = (1 - e^{-t})u(t)$$

Thus, the expression of $v(t)$ is $\boxed{(1 - e^{-t})u(t)}$.

Step 3 of 4

(b)

Consider the differential equation.

$$2\frac{dv}{dt} - v = 3u(t)$$

Simplify further.

$$2\frac{dv}{dt} = 3u + v$$
$$\frac{dv}{3u+v} = \frac{dt}{2}$$

Step 4 of 4

Integrate on both sides with respect to t.

$$\int_{v(0)}^{v(t)} \frac{dv}{3u+v} = \int_0^t \frac{dt}{2}$$
$$\ln(3u+v)\Big|_{v(0)}^{v(t)} = \frac{t}{2}\Big|_0^t$$
$$\ln(3u+v(t)) - \ln(3u+v(0)) = \frac{t-0}{2}$$
$$\ln\left(\frac{3u+v(t)}{3u+v(0)}\right) = \frac{t}{2}$$

Substitute -6 for $v(0)$.

$$\ln\left(\frac{3u+v(t)}{3u-6}\right) = \frac{t}{2}$$
$$\frac{3u(t)+v(t)}{3u(t)-6} = e^{\frac{t}{2}}$$
$$3u(t)+v(t) = 3e^{\frac{t}{2}}u(t) - 6e^{\frac{t}{2}}$$
$$v(t) = 3\left(1-e^{\frac{t}{2}}\right)u(t) - 6e^{\frac{t}{2}}$$

Thus, the expression of $v(t)$ is $\boxed{3\left(1-e^{\frac{t}{2}}\right)u(t)-6e^{\frac{t}{2}}}$.